

selection. In turn, algo providers are required to be more proactive in continually improving their offerings and efficiencies.

- *Reduced Transaction Costs.* Computers are better equipped and faster to react to changing market conditions and unplanned events. They are better capable to ensure consistency between the investment decision and trading instructions, which results in decreased market impact cost, less timing risk, and a higher percentage of completed orders (lower opportunity cost).

Disadvantages

Algorithmic trading has been around only since the early 2000s and it is still evolving at an amazing rate. Unfortunately, algorithms are not the be all and end all for our trading needs. Deficiencies and limitations include:

- Users can become complacent and use the same algorithms regardless of the order characteristics and market conditions simply because they are familiar with the algorithm.
- Users need to continuously test and evaluate algorithms to ensure they are using the algorithms properly and that the algorithms are doing what they are advertised to do. Users need to measure and monitor performance across brokers, algorithms and market conditions to understand what algorithms are most appropriate given the type of market environment.
- Algorithms perform exactly as they are specified, which is nice when the trading environment is what has been expected. However, in the case that unplanned events occur, the algorithm may not be properly trained or programmed for that particular market, which may lead to sub-par performance and higher costs.
- Users need to ensure consistency across the algorithm and their investment needs. Ensuring consistency is becoming increasingly difficult in times where the actual algorithmic trading rule is not as transparent as it could be or when the algorithms are given non-descriptive names that do not provide any insight into what they are trying to do.
- Too many algos and too many names. VWAP, volume weighted average price, is an example of a fairly descriptive algorithmic name and is fairly consistent across brokers. However, an algorithm such as Tarzan is not descriptive and does not provide insights into how it will trade during the day. Investors may need to understand and differentiate between hundreds of algorithms, and keep track of the changes that occur in these codebases. For example, a large institution may use twenty different brokers with five to ten different algorithms each, and with at least half of those names being non-descriptive.